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Autonomic software and system Assignment 1

Task 1

The anti-lock braking system in a car, also known as the ABS, is a system that helps the driver when they must brake hard by pulsating the brake pressure to the pumps in the brakes so that they do not lock up. ABS belongs to the Human assisted (2) level of autonomy since it can act in parallel to a human input, but not without it. In this case ABS needs the human input of pressing the brakes to work. One possible risk that ABS cannot prevent is if the human starts pressing the brakes too late.

The pedestrian detection is a system that can detect pedestrians that are in the path of the vehicle and can either alert the driver or automatically brake if needed. It works based on sensors and works the best at lower speeds. It belongs to the human delegated (3) tier in case it can brake on its own. One possible risk that it cannot prevent is to steer out of the way of a pedestrian if the detection is too late or at a speed where braking is no longer possible in time.

The temperature warning is a system that detects the temperature outside and warns the driver if it is below freezing, which means that there may be ice on the road. This system belongs in the human operated (1) tier since it does not have any actions it can do. One possible risk that it cannot prevent is detecting ice on the road and thus driving more carefully.

Task 2

The MAPE-K autonomic loop is consisted of 5 parts, the monitor, analyser, planner, executor and knowledge. Most of these appear in the first paper, and here are my findings.

The monitor is there by having monitoring for the clusters through the control feedback loop in figure 2 in the paper. The monitoring is done by autonomic controllers that monitor hardware and software utilization metrics per VM.

The analyser is in the same autonomic controller part since they use these monitored metrics to compare what state the utilization is in compared to what it should be at, also called the error, calculated with a PID controller.

The planner is also together with the analyser since the analyser part of MAPE-K is done by the PID controller by taking in data and then deciding what the next move is.

The executor would be the starting and closing of virtual machines by the load balancers since the project is the platform for staying at optimal utilization.

I did not really find a specific knowledge component since I didn't recognize any part that would help exchange knowledge while also saving its own.

For the self-components of configuration, healing, optimization and protection I had a harder time than with the first task.

Self-configuration could be argued to be in the system since it configures the amount of VM: s constantly according to the load and utilization, and it does this without human input. It can also adapt to various unpredictable conditions regarding the utilization so I would say that the self-configuration is represented well.

The self-healing however is harder to define since the system does not mention any types of fail safes if it crashes.

Self-optimization is what the whole system is about, and it strives to always improve the system operation towards the calculated ideal. It constantly tries to improve the performance of the system by changing the amount of VM: s according to the utilization.

Self-protection is a lot like self-healing since there is no real mention of a detection system for it or any other types of security features.

Well since I have also made a load balancer with a scaling rule in a cloud course I remember that I had a similar result as in figure 6 (not as good of course but similar), since the utilization in my system also spiked at the start since it takes a while for the load balancer to start up another machine after the initial spike, which in my case led to the first VM having 100% utilization, while theirs shows a ~70% spike. This is the only thing I could see that could clearly improve, and maybe that could be done with some type of pre-emptive method that would recognise when a potential spike is coming and launch VM: s to prepare for it.